SPHOORTHY ENGINEERING COLLEGE

An Autonomous Institution Approved by AICTE & Accredited with NAAC Grade 'A'
Department of Examination & Academic Affairs · Hyderabad — 501510

Computer Networks: OSI/TCP-IP Protocols, Routing & Congestion Control

Program / Dept: B.Tech / MBA — CSE	**Course Code / Title:** Computer Networks (CS502PC)
Academic Regulation: Autonomous (SR23 / SR24)	**Curriculum Semester:** Semester 5 (Year 3)
Credits / Contact Hours: 3 Credits · 48 Lecture Hours	**Evaluation Scheme:** Continuous Internal (40M) + End Sem (60M)

Course Overview & Pedagogical Objectives:

Layered protocol stacks, Data link framing & CRC, Sliding Window protocols, Routing algorithms (OSPF, BGP), TCP Reno/Tahoe, and Application layer protocols (DNS, HTTP/2). The primary objective of this course is to equip students with sound theoretical rigor, analytical problem-solving competencies, and real-world system engineering capabilities demanded by premier technological organizations and competitive graduate benchmarks.

Course Outcomes (COs) & Bloom's Taxonomy Mapping:

CO ID	Course Learning Outcome Description	Bloom's Level
CO 1	Understand and formulate fundamental concepts of Unit 1: Network Physical & Data Link Layer.	L2 Understand
CO 2	Analyze algorithms, mechanisms, and models in Unit 2: Medium Access Control (MAC).	L4 Analyze
CO 3	Design and evaluate robust architectures for Unit 3: Network Layer & Routing.	L5 Evaluate
CO 4	Implement and optimize operational pipelines in Unit 4: Transport Layer.	L3 Apply
CO 5	Integrate state-of-the-art frameworks in Unit 5: Application Layer & Security for industrial solutions.	L6 Create

Comprehensive Module Index:

Unit	Title & Core Concepts	Page
Unit 1	**Unit 1: Network Physical & Data Link Layer**: OSI 7-layer reference model, TCP/IP architecture, Framing, Flow control (Stop-and-Wait, Go-Back-N, Selective R...	Page 2
Unit 2	**Unit 2: Medium Access Control (MAC)**: ALOHA (Pure and Slotted), CSMA/CD (Ethernet IEEE 802.3), CSMA/CA (Wi-Fi IEEE 802.11), and Switching methods (V...	Page 3
Unit 3	**Unit 3: Network Layer & Routing**: IPv4 addressing, Subnetting, CIDR, IPv6 header format, Distance Vector Routing (Bellman-Ford), Link State Rout...	Page 4
Unit 4	**Unit 4: Transport Layer**: User Datagram Protocol (UDP), Transmission Control Protocol (TCP), 3-way handshake, Flow control, TCP Congesti...	Page 5
Unit 5	**Unit 5: Application Layer & Security**: Domain Name System (DNS), HTTP/1.1 vs HTTP/2, SMTP, POP3, IMAP, SSL/TLS handshake overview, and symmetric/asym...	Page 6

UNIT 1: NETWORK PHYSICAL & DATA LINK LAYER • *Core Theory & Practice Guide*

1.1 Technical Exposition & Conceptual Foundations

The study of unit 1: network physical & data link layer forms an indispensable cornerstone of Computer Networks (CS502PC). Specifically, OSI 7-layer reference model, TCP/IP architecture, Framing, Flow control (Stop-and-Wait, Go-Back-N, Selective Repeat), Error detection (CRC, Checksum). Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

1.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 1 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

1.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 1
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

1.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 1 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 1. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 1 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 1 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 2: MEDIUM ACCESS CONTROL (MAC) • *Core Theory & Practice Guide*

2.1 Technical Exposition & Conceptual Foundations

The study of unit 2: medium access control (mac) forms an indispensable cornerstone of Computer Networks (CS502PC). Specifically, ALOHA (Pure and Slotted), CSMA/CD (Ethernet IEEE 802.3), CSMA/CA (Wi-Fi IEEE 802.11), and Switching methods (Virtual circuit vs Datagram). Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

2.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 2 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

2.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 2
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

2.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 2 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 2. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 2 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 2 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 3: NETWORK LAYER & ROUTING • *Core Theory & Practice Guide*

3.1 Technical Exposition & Conceptual Foundations

The study of unit 3: network layer & routing forms an indispensable cornerstone of Computer Networks (CS502PC). Specifically, IPv4 addressing, Subnetting, CIDR, IPv6 header format, Distance Vector Routing (Bellman-Ford), Link State Routing (Dijkstra/OSPF), and BGP routing. Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

3.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 3 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

3.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 3
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

3.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 3 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 3. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 3 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 3 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 4: TRANSPORT LAYER • *Core Theory & Practice Guide*

4.1 Technical Exposition & Conceptual Foundations

The study of unit 4: transport layer forms an indispensable cornerstone of Computer Networks (CS502PC). Specifically, User Datagram Protocol (UDP), Transmission Control Protocol (TCP), 3-way handshake, Flow control, TCP Congestion Control algorithms (AIMD, Slow Start). Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

4.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 4 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

4.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 4
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

4.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 4 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 4. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 4 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 4 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

UNIT 5: APPLICATION LAYER & SECURITY • *Core Theory & Practice Guide*

5.1 Technical Exposition & Conceptual Foundations

The study of unit 5: application layer & security forms an indispensable cornerstone of Computer Networks (CS502PC). Specifically, Domain Name System (DNS), HTTP/1.1 vs HTTP/2, SMTP, POP3, IMAP, SSL/TLS handshake overview, and symmetric/asymmetric cryptographic principles. Modern applications require deep awareness of edge conditions, invariant properties, and trade-offs between computational throughput, latency overhead, memory footprint, and architectural scalability.

In university examinations and practical engineering systems, solutions must be formulated with mathematical clarity. Each sub-mechanism must satisfy safety, liveness, and fault-tolerance constraints while adhering to standard autonomous syllabus requirements.

5.2 Architectural Principles & Comparative Analysis

Component / Topic	Key Mechanism & Principle	Operational Trade-off
Theoretical Model	Formal representation of Unit 5 entities and lifecycle stages.	Higher mathematical abstraction vs implementation friction.
Algorithmic Engine	State machine transitions, recurrence guarantees, and verification steps.	Time complexity bounds vs memory consumption.
Real-World Systems	Deployment in enterprise distributed systems and contemporary toolchains.	Fault-tolerant redundancy vs network overhead.

5.3 Practical Formulation & Execution Workflow

```
// Standard Implementation Outline & Pseudocode for Unit 5
1. Initialize system data structures and state variables under initial constraints.
2. Execute continuous invariant checking across state boundaries.
3. Apply optimization bounds: O(log N) average time or amortized linear bounds.
4. Catch boundary failure cases (underflow, overflow, concurrency race, null references).
5. Commit verified state or invoke transactional rollback upon validation failure.
```

5.4 Autonomous End-Semester Examination Question Bank

Type	Question & Expected Technical Answer Outline	Marks
Part A	**Q1:** Define the primary objective of Unit 5 and state two essential real-world applications. • *Expected: Exact definition, mathematical constraint formulation, and dual industry examples.*	**2 M**
Part A	**Q2:** Differentiate between static vs dynamic formulations of key mechanisms in Unit 5. • *Expected: 2-column comparison table covering memory, allocation time, and failure recovery.*	**2 M**
Part B	**Q3:** Explain the complete working architecture of Unit 5 with neat functional block diagrams and step-by-step algorithmic trace. • *Expected: Labeled diagram (3M), mathematical derivation (3M), algorithm steps (2M), complexity analysis (2M).*	**10 M**
Part B	**Q4:** Analyze the worst-case, best-case, and average-case performance bounds for processes in Unit 5 with worked numerical example. • *Expected: Step-by-step problem solution, boundary checks, and conclusive optimality proof.*	**10 M**

Standard Recommended Textbooks & Digital References:

1. *Primary Curriculum Standard Textbook* — Pearson Higher Education / McGraw-Hill Education (Latest Edition).
2. *Autonomous Reference Syllabus Blueprint* — Sphoorthy Engineering College Academic Council.
3. *NPTEL & IEEE Computer Society Guidelines* — Open Educational Resources & Peer-reviewed Publications.